PROFESSIONAL, TECHNICAL AND ANALYTICAL SUPPORT TO NAVY ELECTRONIC WARFARE

1.0 INTRODUCTION AND GENERAL INFORMATION:

1.1 INTRODUCTION

This Performance Work Statement (PWS) supports Naval Information Warfighting Development Center (NIWDC) warfighting capabilities with specific focus towards Electronic Warfare Data (EWD)production (using the Navy Information Operations Database (NIODB)) currently operated on an Amazon Web Service (AWS) Impact Level (IL6) cloud hosting environment. NIODB is a Navy approved source of parametric, and order of battle data used in the development of Electronic Warfare (EW) System Libraries. Aiding the warfighter with a single source of authoritative EW data, it improves threat and own force identification, enables Anti-Ship Missile Defense, enhances interoperability, and improves mission planning through improved threat data fidelity. OPNAV Instruction 3430.23D provides policy for United States Fleet Forces Command (USFFC) and Naval Information Forces (NAVIFOR) execution of the Electronic Warfare Reprogrammable Systems Support Program. NIWDC EWD is NAVIFOR designated EW Functional Data Manager and is the primary organization for this support.

1.2 TYPE OF CONTRACT

The Government intends to award an Indefinite Delivery/Indefinite Quantity contract with Firm-Fixed Price provisions for services. Individual task orders will specify positions.

1.3 ORDERING PERIOD

The resultant Indefinite Delivery, Indefinite Quantity (IDIQ) contract will consist of the following five year base ordering period and will contain FAR 52.217-8 – Option to Extend Services to allow for a six- month extension of the ordering period if circumstances necessitate the exercise of this option. The ordering periods are as follows:

Base Ordering Period:	28 March 2027 – 27 March 2032
FAR 52.217-8 Option Ordering Period:	28 March 2032 – 27 September 2032

1.4 PERIOD OF PERFORMANCE

Base Year:	28 March 2027 - 27 March 2028
Ordering Year I:	28 March 2028 - 27 March 2029
Ordering Year II:	28 March 2029 - 27 March 2030
Ordering Year III:	28 March 2030 - 27 March 2031
Ordering Year IV:	28 March 2031 - 27 March 2032
FAR 52.217-8:	28 March 2032 - 27 September 2032

NOTE: The performance period of individual task orders will be established at the time of task order issuance.

1.5 PLACE OF PERFORMANCE/WORKING HOURS

Performance will occur at the following locations:

- Naval Information Warfare Training Group Norfolk, 1247 West C Street, Building U-132, Naval Station, Norfolk, VA. 23511
- Vendor company accredited facility/corporate office upon COR approval
- Working hours are Monday – Friday, 7:30 am – 4:00 pm except Federal Holidays. The expectation is an 8-hour day.

1.6 OFFICE SPACE

The Government will provide office space, desks, chairs, and common office supplies. In addition to office space, each Contractor employee shall be provided with access to government computers and telephones, but not cellular phones, for official use only. Under no circumstances will Contractor-provided personal computers, or any personal electronic devices (e.g., cell phones, tablets, Fitbits), be connected to any Government IT system, including but not limited to the Navy/Marine Corps Intranet (NMCI). Contractor personnel shall comply with all DOD and Navy IT regulations, policies and procedures including those concerning internet usage and cybersecurity.

For contract personnel working at the vendor accredited facility/corporate office, the vendor will furnish material/workspace as necessary to perform assigned tasks.

1.7 HOLIDAYS:

The contractor is not required to provide services on these days:

New Year's Day
Martin Luther King Jr.'s Birthday
President's Day
Memorial Day
Juneteenth Day
Independence Day
Labor Day
Columbus Day
Veteran's Day
Thanksgiving Day
Christmas Day

2.0 SCOPE:

This PWS encompasses support for NIWDC Missions, Functions, and Tasks (MF&T) related to the mission areas of Navy Electronic Warfare Functional Data Management and Navy Information Operations Database Support. This PWS provides services and support that includes both short-term and annual support contracts in some or all of the following EW tasks.

3.0 TASK DESCRIPTION:
The following tasks describe NIWDC services and support required under this contract:

Labor Category	Required Support & Qualifications	CSWF Level	Quantity of Personnel
*Technical SIGINT Data & Electronic Warfare Requirements Analysis Support**:** **Subject Matter Expert II**	3.1.1 and 3.1.2	N/A	6
Technical SIGINT Data & Electronic Warfare Requirements Analysis Support: **Subject Matter Expert I**	3.1.3 and 3.1.4	N/A	2
Electronic Warfare Reprogrammable Systems Library Analysis Support: **Subject Matter Expert II**	3.2.1 and 3.2.2	N/A	3
Electronic Warfare Reprogrammable Systems Library Analysis Support: **Subject Matter Expert I**	3.2.3 and 3.2.4	N/A	1
Navy Information Operations Database (NIODB) /Software Development and Sustainment Support: **Database Administrator IV	3.3.1 and 3.3.2	Intermediate	11
Service Administrator Support: **Windows Server Administrator**	3.4.1 and 3.4.2	Intermediate	3
Service Administrator Support: **Linux Sever Administrator**	3.4.3 and 3.4.4	Intermediate	3
Cloud Engineer Support: **Cloud Engineer II**	3.5.1 and 3.5.2	Intermediate	2
Data Scientist	3.6.1 and 3.6.2	Intermediate	2

* One (1) Technical SIGINT Data & Electronic Warfare Requirements Analysis Support**:** **Subject Matter Expert II** will be designated as Key Personnel.

One (1) Navy Information Operations Database (NIODB) /Software Development and Sustainment Support: **Database Administrator IV will be designated as Key Personnel.

The Government reserves the right to review the resume and request validation of certifications and experience of contractor staff designated as Key Personnel.

3.1 Technical SIGINT Data & Electronic Warfare Requirements Analysis Support:

Estimated FTE:
1. Subject Matter Expert II (Key Person): 1912 hrs

2. Subject Matter Expert II: 1912 hrs
3. Subject Matter Expert II: 1912 hrs
4. Subject Matter Expert II: 1912 hrs
5. Subject Matter Expert II: 1912 hrs
6. Subject Matter Expert II: 1912 hrs
7. Subject Matter Expert I: 1912 hrs
8. Subject Matter Expert I: 1912 hrs

3.1.1 Subject Matter Expert II Functional Responsibility:

Technical Signals Intelligence (SIGINT) & Electronic Warfare Data and System Requirements analysis, technical guidance for data management/interoperability, database design and development, data standardization and maintenance, data migration, content quality assurance support of the Navy Information Operations Database (NIODB), and development of technical reports.

This support shall include:

- Conduct detailed analysis of complex communication & non-communication waveforms to determine pulse repetition intervals (PRI), pulse duration, and scanning characteristics.
- Liaison with Navy and DoD Electronic Warfare system engineers to document, interpret, and analyze current and fielded EW system/community data requirements.
- Research, identify, and validate authoritative information sources to support documented and validated training system/community requirements.
- Provide Subject Matter Expertise in Technical SIGINT data analysis (i.e. Electronic Warfare Integrated Reprogramming (EWIR), Combined Emitter Database (CED)) and National Intelligence Product (ONI, NASIC, NGIC, MSIC, NGA, DIA, NSA) analysis in support of Electronic Warfare (EW) systems.

3.1.2 Subject Matter Expert II Qualification Required:

The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- 10 years demonstrated experience in the field of Technical Electronic Intelligence (TECHELINT) data analysis and authoritative national data sources to include EWIR, and CED
- Demonstrated experience in the field of Electronic Warfare systems, RADAR threat parameter analysis, threat/friendly order of battle (OOB) analysis, Specific Emitter Identification (SEI), and authoritative national data sources to include ONI, NASIC, NGIC, MSIC, NGA, DIA, and NSA.
- Demonstrated experience in the use of data query and visualization tools to include CA ERwin Data Modeling Software, MS Excel, and Oracle SQL Developer.
- Comprehensive knowledge of data rationalization and standardization efforts to document and cross-correlate EW system data requirements in addition to interpreting highly technical Database Description Documents (DBDD) and System Requirements Specifications (SRS).

- Ability to effectively communicate orally and in writing to produce finished products, liaise with current and potential customers, prepare and present briefing documents, charts, graphs, and presentations.

3.1.3 Subject Matter Expert I Functional Responsibility:
Technical Signals Intelligence (SIGINT) & Electronic Warfare Data and System Requirements analysis, technical guidance for data management/interoperability, database design and development, data standardization and maintenance, data migration, content quality assurance support of the Navy Information Operations Database (NIODB), and development of technical reports.

This support shall include:

- Liaison with Software Developers to produce, assess, and validate unique EW data products developed from NIODB in support of Navy and DoD EW systems/community partners.
- Research, development, recommendation, and documentation of Quality Assurance processes, data analysis of derivative database product development, and QA verification tasks related to NIODB.
- Create detailed technical papers recommending and describing QA processes that will ensure development of intuitive, precise, user friendly, easily integrated, web enabled QA tools.

3.1.4 Subject Matter Expert I Qualification Required:
The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- Demonstrated experience in the field of Electronic Warfare systems, RADAR threat parameter analysis, threat/friendly order of battle (OOB) analysis, Specific Emitter Identification (SEI), and authoritative national data sources to include ONI, NASIC, NGIC, MSIC, NGA, DIA, and NSA.
- Demonstrated experience in the use of data query and visualization tools to include CA ERwin Data Modeling Software, MS Excel, and Oracle SQL Developer.
- Demonstrated experience in Own Force Data (OFD) platform/parametric analysis to support production of validated data in the NIODB and derivative products.
- Demonstrated experience in North Atlantic Treaty Organization (NATO) platform/parametric analysis to support production of releasable validated data into the Joint Electronic Warfare Centers NATO database, SCORPIO.
- Ability to effectively communicate orally and in writing to produce finished products, liaise with current and potential customers, prepare and present briefing documents, charts, graphs, and presentations

3.2 Electronic Warfare Reprogrammable Systems Library Analysis Support:

Estimated FTE:
1. Subject Matter Expert II: 1912 hrs
2. Subject Matter Expert II: 1912 hrs
3. Subject Matter Expert II: 1912 hrs
4. Subject Matter Expert I: 1912 hrs

3.2.1 Subject Matter Expert II Functional Responsibility:
The Contractor shall provide engineering and technical support, assistance, and training for EW Reprogrammable Libraries, also known as “Threat Databases” in support of AN/SLQ-32 Anti-Ship Missile Defense (ASMD) , training databases and AN/BLQ-10B subsurface Electronic Surveillance (ES) threat libraries. Services include: Interpreting intelligence information, performing multi-int analysis, assigning threat levels, knowledge of geo-area filtering, performing parametric mode combining, implementing system white paper rules, writing/modifying technical documents, and researching ELINT/Order of Battle (OOB) platform threat data from national agency (i.e. ONI/NASIC/NGIC/MIDB/DIA) sources. Perform analysis using the Navy Information Operations Database (NIODB), (CED), and Electronic Warfare Integrated Reprogramming Database (EWIRDB). The library build process is continuous throughout the year.

This support shall include:

- Research, identify, and validate authoritative information sources to support production of sensor parameter data in NIODB and derivative products.
- Conduct detailed analysis of complex non-communication waveforms to determine pulse repetition intervals (PRI), pulse duration, and scanning characteristics to support development of SLQ-32(V) Tactical Trainer libraries.

3.2.2 Subject Matter Expert II Qualification Required:
The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- 10 years demonstrated experience in the field of Technical ELINT (TECHELINT) data analysis and authoritative national data sources to include NGES, EWIR, and CED.
- Demonstrated experience in the field of Electronic Warfare systems, RADAR threat parameter analysis, threat/friendly order of battle (OOB) analysis, Specific Emitter Identification (SEI), and authoritative national data sources to include ONI, NASIC, NGIC, MSIC, NGA, DIA, and NSA.
- Demonstrated experience with the SLQ-32(V) and NCTE/LVC is highly desired but not required.
- Demonstrated operational EW experience with the Navy’s tactical training libraries to include proven experience with tactical training simulation systems
- Demonstrated experience in the use of data query and visualization tools such as MS Excel.

- Ability to effectively communicate orally and in writing to produce finished products, liaise with current and potential customers, prepare and present briefing documents, charts, graphs, and presentations.
- Demonstrated experience in ELINT analysis to support production of sensor parameter data in NIODB and derivative products. Specific expertise utilizing authoritative data sources (i.e., DIA EWIR/NSA CED) required.
- Demonstrated experience in Surface and Subsurface Electronic Warfare (EW) threat libraries to support development of SLQ-32(V) Anti-Ship Missile Defense (ASMD) threat libraries and AN/BLQ-10B Electronic Surveillance threat libraries. Operational SLQ-32(V) and BLQ-10B experience highly desired but not required.
- Proven experience in TECHELINT analysis to support development of the AN/SLQ-32(V) Tactical Trainer libraries.

3.2.3 Subject Matter Expert I Functional Responsibility:

The Contractor shall provide engineering and technical support, assistance, and training for EW Reprogrammable Libraries, also known as “Threat Databases” in support of AN/SLQ-32 Anti-Ship Missile Defense (ASMD) , training databases and AN/BLQ-10B subsurface Electronic Surveillance (ES) threat libraries. Services include: Interpreting intelligence information, performing multi-int analysis, assigning threat levels, knowledge of geo-area filtering, performing parametric mode combining, implementing system white paper rules, writing/modifying technical documents, and researching ELINT/Order of Battle (OOB) platform threat data from national agency (i.e. ONI/NASIC/NGIC/MIDB/DIA) sources. Perform analysis using the Navy Information Operations Database (NIODB), (CED), and Electronic Warfare Integrated Reprogramming Database (EWIRDB). The library build process is continuous throughout the year.

This support shall include:

- Conduct Worldwide Order Of Battle Analysis to support production of data in the NIODB and derivative products utilizing authoritative data sources (i.e., ONI/NASIC/NGIC/MSIC).
- Research, identify and validate detailed parametric and platform characteristics and performance data in support of development of Surface and Subsurface Electronic Warfare (EW) threat libraries to include the SLQ-32(V) Anti-Ship Missile Defense (ASMD) threat libraries and AN/BLQ-10B Electronic Surveillance threat libraries.

3.2.4 Subject Matter Expert I Qualification Required

The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- Demonstrated operational experience with the Navy’s Surface Electronic Warfare Tactical Training libraries.
- Demonstrated experience in the use of data query and visualization tools such as MS Excel.

- Ability to effectively communicate orally and in writing to produce finished products, liaise with current and potential customers, prepare and present briefing documents, charts, graphs, and presentations.
- Demonstrated experience in Worldwide Order Of Battle Analysis to support production of data in the NIODB and derivative products. Specific expertise utilizing authoritative data sources (i.e., ONI/NASIC/NGIC/MSIC) required.
- Operational experience with the AN/SLQ-32(V) and NCTE/LVC experience highly desired but not required.

3.3 Navy Information Operations Database (NIODB) /Software Development and Sustainment Support:

Estimated FTE:

1. Database Administrator IV (Key Person): 1912 hrs
2. Database Administrator IV: 1912 hrs
3. Database Administrator IV: 1912 hrs
4. Database Administrator IV: 1912 hrs
5. Database Administrator IV: 1912 hrs
6. Database Administrator IV: 1912 hrs
7. Database Administrator IV: 1912 hrs
8. Database Administrator IV: 1912 hrs
9. Database Administrator IV: 1912 hrs
10. Database Administrator IV: 1912 hrs
11. Database Administrator IV: 1912 hrs

3.3.1 Functional Responsibility:

The Contractor shall provide engineering and technical support, assistance, and training for data management, interoperability, database design and development, data standardization and maintenance, data migration, web application software design and development, and EW data analysis/library content development in support of the Navy Information Operations Database (NIODB

This support shall include:

- o Data architecture development for NIODB support to legacy and emerging systems
- o Support current programming languages (i.e., C#, JQuery, Javascript, JSON, Angular) while advising of new technologies and industry trends.
- o Support conversion of legacy data formats to modern/next generation (.i.e., .CSV,.XML,JSON)
- o Support maintenance, security, and documentation of software installation and configuration base line.

- Support development and sustainment of NIODB workflow and quality assurance solutions in support of all IW Data Management functions/products.
- Support network configuration managers, system administrators, and information system security professionals to understand and document development environment requirements.
- Liaison with National and Service Organizations for incorporating new database formats into the Navy's authoritative database.
- Support establishment and maintenance of an alternate NIODB site to sustain disaster recovery and continuity of operations.
- Support Testing, verification, and validation of data products and data management applications.
- Support maintenance, development, documentation and sustainment of Surface Electronic Warfare training databases/libraries.
- Support maintenance, development, documentation and sustainment of Next Generation BLQ-10B subsurface Electronic Support (ES) libraries
- Develop of Plan of Action and Milestones (POAM) and related documentation to support Fleet requirements and related IO capabilities.
- Support project management employing Agile and sprints as an iterative approach to deliver products/services faster and with less re-work.

3.3.2 Qualification Required:

The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- 5+ years demonstrated experience with relational database management systems development with a preference for Oracle, including Simple Query Language (SQL), PL/SQL, and/or T-SQL
- Demonstrated experience in software application/database design to support development and maintenance of the SIPR NIODB. NIODB is a Microsoft Web Forms / MVC front-end web application supported by a back-end Oracle Enterprise RDBMS.
- Demonstrated experience in the field of software development with specific prior performance in .NET desktop & web application development.
- Demonstrated experience in the field of software development with specific prior performance eXtensable Markup Language (XML) including the use of XML schemas and eXtensible (XSLT) Stylesheet Language (XSLT) to support parsing and import of external data products to an Oracle Relational Database Management System (RDBMS).
- Demonstrated experience in software application/database design to support production of Subsurface EW threat libraries (i.e., SLQ-32, BLQ-10B).
- Demonstrated experience in software application/database design to support development/maintenance of the NIODB Desktop (stand-alone). The NIODB Desktop is a Microsoft .NET WPF application utilizing .XML data stores.
- Contractors must hold the appropriate certification per Defense Federal Acquisition Regulation Supplement (DFARS) 48 Code of Regulations (CFR) Parts 239 and 252 Record Identification Number (RIN) 0750-AF52 DFARS: Information Assurance Contractor Training and Certification (DFARS Case 2006-D023). Candidate(s) will

require privileged access to development and production systems and must be DOD-8140 Cyber Security Workforce (CSWF) Certified with specific certification of Security+ as a day one requirement.

3.4 Server Administrator Support:

Estimated FTE:
1. Windows Server Administrator: 1912 hrs
2. Windows Server Administrator: 1912 hrs
3. Windows Server Administrator: 1912 hrs
4. Linux Server Administrator: 1912 hrs
5. Linux Server Administrator: 1912 hrs
6. Linux Server Administrator: 1912 hrs

3.4.1 Windows Server Administrator Functional Responsibilities:

The contractor shall provide technical support, configuration, operation, security, availability, backup, and updates for Windows servers supporting, but not limited to: web servers, database servers, developer and analyst workstations, and their associated images. Support Services shall include:

- The Systems Administrator (SA) shall manage the software/application, security, and backups of the Windows servers/workstations to support all NIODB services. The activities include (but are not limited to): ensuring the servers and all installed software are compliant with DoD STIGs, performing access management in accordance with least privilege, updating and patching server hardware and software, reviewing audit logs for security and performance and performing back-ups.
- The SA shall be responsible for assisting in the development of a disaster recovery plan to ensure that NIODB services are continuously available with minimal to no interruption and maximum transparency to users.
- The SA shall provide the government with recommended solutions to meet changing requirements and ensure the NIODB services are equipped with the most up to date relevant tools.

3.4.2 Windows Server Administrator Qualification Required:

The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- Contractors must hold the appropriate certification per Defense Federal   Acquisition Regulation Supplement (DFARS) 48 Code of Federal Regulations   (CFR) Parts 239 and 252 Record Identification Number (RIN) 0750-AF52 DFARS: Information Assurance Contractor Training and Certification (DFARS Case 2006-D023). Candidate(s) will require privileged access to Windows/Linux systems. Candidates must be DOD-8140 Cyber Security Workforce (CSWF) Certified with specific certification of Security+ as a day one requirement. Within six months of assignment of duties, server administrators shall verify attained certification in currently supported Windows certification or other appropriate certification identified by the government.

- Bachelors in Systems Administration, Computer Science, Information Technology, or related field experience preferred.
- Certification in current Windows to support assigned duties.
- 3+ years demonstrated Windows systems administration experience to include management of web and database servers.
- Strong knowledge of backup and restoral procedures preferred.

3.4.3 Linux Server Administrator Functional Responsibilities:

The contractor shall provide technical support, configuration, operation, security, availability, backup, and updates for Linux servers supporting, but not limited to: web servers, database servers, developer and analyst workstations, and their associated images. Support Services shall include:

- The Systems Administrator (SA) shall manage the software/application, security, and backups of the Linux servers/workstations to support all NIODB services. The activities include (but are not limited to): ensuring the servers and all installed software are compliant with DoD STIGs, performing access management in accordance with least privilege, updating and patching server hardware and software, reviewing audit logs for security and performance and performing back-ups.
- The SA shall be responsible for assisting in the development of a disaster recovery plan to ensure that NIODB services are continuously available with minimal to no interruption and maximum transparency to users.
- The SA shall provide the government with recommended solutions to meet changing requirements and ensure the NIODB services are equipped with the most up to date relevant tools.

3.4.4 Linux Server Administrator Qualification Required:

The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- Contractors must hold the appropriate certification per Defense Federal Acquisition Regulation Supplement (DFARS) 48 Code of Federal Regulations (CFR) Parts 239 and 252 Record Identification Number (RIN) 0750-AF52 DFARS: Information Assurance Contractor Training and Certification (DFARS Case 2006-D023). Candidate(s) will require privileged access to Linux systems. Candidates must be DOD-8140 Cyber Security Workforce (CSWF) Certified with specific certification of Security+ as a day one requirement. Within six months of assignment of duties, server administrators shall verify attained certification in currently supported Linux certification or other appropriate certification identified by the government.
- Bachelors in Systems Administration, Computer Science, Information Technology, or related field experience preferred.
- Certification in current Linux to support assigned duties.

- 3+ years demonstrated Linux systems administration experience to include management of web and database servers.
- Strong knowledge of backup and restoral procedures preferred.

3.5 Cloud Engineer Support:

Estimated FTE:
1. Cloud Engineer II: 1912 hrs
2. Cloud Engineer II: 1912 hrs

3.5.1 Functional Responsibilities:

The contractor shall provide a qualified Cloud Engineer to design, implement, and manage secure cloud infrastructure in support of NIWDC mission objectives. The primary responsibilities include, but are not limited to, the following:

- Cloud Infrastructure Development: Design, build, and deploy secure, scalable, and resilient cloud-native and hybrid cloud solutions in accordance with DoD and DoW architectural standards. This includes leveraging Infrastructure as Code (IaC) with tools such as Terraform or CloudFormation.
- System Administration & Automation: Perform system administration for cloud-based services, including virtual machines, containers (e.g., Kubernetes), and platform services. The contractor shall automate routine operational tasks, monitoring, and security functions using scripting languages (e.g., Python, Bash).
- Security & Compliance: Implement and maintain security controls to ensure compliance with IL6 security requirements. The contractor shall be responsible for configuring identity and access management (IAM), network security groups, encryption, and logging/monitoring solutions to prevent, detect, and respond to threats.
- Continuous Integration/Continuous Deployment (CI/CD): Develop and manage CI/CD pipelines to automate the testing and deployment of applications and infrastructure updates, ensuring minimal downtime and rapid delivery of capabilities.
- Troubleshooting & Optimization: Provide Tier III support for cloud infrastructure. The contractor will diagnose and resolve complex system issues, optimize performance, and monitor resource utilization to ensure cost-efficiency and operational effectiveness.

3.5.2 Qualification Required:

The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- Bachelor's Degree in Computer Science, Information Technology, Engineering, or a related technical field.
- A minimum of five (5) years of hands-on experience in cloud engineering, with at least two (2) years focused on deploying solutions in AWS, Azure, or Google Cloud. Experience in a government or military environment is highly preferred.

- Proficiency with IaC tools (Terraform, CloudFormation).- Strong experience with containerization (Docker, Kubernetes).- Advanced scripting skills (Python, Bash, PowerShell).- Deep understanding of cloud networking and security principles (VPCs, IAM, Security Groups).
- Must hold at least one professional-level cloud certification (e.g., AWS Certified Solutions Architect Professional, Azure Solutions Architect Expert) and a DoD 8570 IAT Level II certification (e.g., Security+).
- Contractors must hold the appropriate certification per Defense Federal Acquisition Regulation Supplement (DFARS) 48 Code of Federal Regulations (CFR) Parts 239 and 252 Record Identification Number (RIN) 0750-AF52 DFARS: Information Assurance Contractor Training and Certification (DFARS Case 2006-D023). Candidate(s) will require privileged access to development and production systems and must be DOD-8140 Cyber Security Workforce (CSWF) Certified with specific certification of Security+ as a day one requirement.

3.6 Data Scientist:

Estimated FTE:
1. Data Scientist: 1912 hrs
2. Data Scientist: 1912 hrs

3.6.1 Functional Responsibilities:

The Data Scientist will provide expertise in data analytics and machine learning to exploit, analyze, and derive insights from complex EW and Signals Intelligence (SIGINT) data. The core mission is to enhance situational awareness, optimize EW systems, and provide data-driven support to the warfighter.

This support shall include:

- Analyze large-scale, high-velocity datasets from various sources, including RF sensors, national-level SIGINT, and operational mission data. Identify patterns, anomalies, and correlations to characterize the electromagnetic environment.
- Apply advanced signal processing techniques to raw sensor data (e.g., I/Q data) to extract relevant features. Develop novel features for signal classification, identification, and parameter estimation.
- Design, develop, and validate machine learning (ML) and deep learning (DL) models for tasks such as automatic signal detection, classification (e.g., Specific Emitter Identification - SEI), and threat behavior prediction.
- Process and refine data to generate and update Mission Data Files (MDFs) for EW systems, ensuring platforms have the most accurate and relevant threat signatures and countermeasure parameters.
- Fuse EW data with all-source intelligence (e.g., GEOINT, OSINT, HUMINT) to build a comprehensive Electronic Order of Battle (EOB) and provide enriched context for operational planning.

- Develop and use M&S environments to test and evaluate the performance of EW algorithms and tactics against simulated threat systems and congested electromagnetic environments.
- Clearly communicate complex technical findings, model performance, and operational recommendations to diverse audiences, from system engineers to operational commanders.

3.6.2 Qualification Required:
The following knowledge, skills and abilities should be considered the minimum requirements for this position:

- Bachelor's degree in a STEM field such as Electrical Engineering, Computer Science, Physics, or Mathematics is required. A Master's or Ph.D. is highly desired.
- Demonstrated a minimum of 5 years of professional experience in data science, including the development of predictive models, statistical analysis, and the deployment of machine learning algorithms in production environments."
- Programming: Proficiency in Python (with libraries like SciPy, NumPy, Pandas, Scikit-learn, PyTorch/TensorFlow) and MATLAB.• Signal Processing: Deep understanding of digital signal processing (DSP), Fourier analysis, and RF principles.• Database Management: Experience with SQL and NoSQL databases for managing large volumes of structured and unstructured data.• Big Data Technologies: Familiarity with distributed computing environments (e.g., Hadoop, Spark) is preferred.
- Knowledge of machine learning and deep learning techniques is necessary for predictive modeling.
- EW Principles: In-depth knowledge of Electronic Attack (EA), Electronic Protection (EP), and Electronic Support (ES).
    - Threat Systems: Understanding of modern radar, communication systems, and associated weapon systems.
    - SIGINT Process: Familiarity with the Tasking, Collection, Processing, Exploitation, and Dissemination (TCPED) cycle.
- Contractors must hold the appropriate certification per Defense Federal Acquisition Regulation Supplement (DFARS) 48 Code of Federal Regulations (CFR) Parts 239 and 252 Record Identification Number (RIN) 0750-AF52 DFARS: Information Assurance Contractor Training and Certification (DFARS Case 2006- D023). Candidate(s) will require privileged access to development and production systems and must be DOD-8140 Cyber Security Workforce (CSWF) Certified with specific certification of Security+ as a day one requirement.

4.0 TRAVEL

The Contractor shall be required to travel in the performance of the tasking detailed in this PWS. Travel costs for transportation, lodging, meals and incidental expenses are allowable if incurred by contractor personnel on official business. Travel related costs shall be reasonable. Costs for transportation may be based on mileage rates, actual costs incurred, or a combination thereof, provided the method used results in a reasonable charge. Costs incurred for lodging, meals and incidental expenses shall be considered reasonable and allowable only to the extent they do not exceed on a daily basis the maximum per diem rates in effect at the time of travel.

The contractor shall use Norfolk, Virgina as the point of origin for travel pricing.

Travel Requirements

Location	# of Travelers	# of Trips	# Days
Point Mugu, CA	2	2	5
Dayton, OH	2	2	4
Dahlgren, VA	2	4	2
Huntsville, AL	2	2	4
Fairmont, WV	2	2	2
San Antonio, TX	2	2	2
Honolulu, HI	1	1	5
Everett, WA	1	1	5
San Diego, CA	1	1	5

5.0 CONTUNITY OF OPERATIONS

5.1 TRANSISTION PLAN
The Contractor shall prepare and maintain a transition plan that ensures a smooth transition from contract start date to full operational status (incoming turnover period) and a smooth transition from current contract performance to performance by a different Contractor or by the Government in a follow-on period (outgoing turnover). The contractor shall provide a Transition Plan that clearly details an approach/plan for the turnover of services defined in the PWS.

5.2 INCOMING TURNOVER PERIOD
During the turnover period, the successor contractor shall prepare to assume full responsibility for all areas of the incumbent functions in accordance with the terms and conditions of the contract. The successor contractor shall meet with the incumbent and discuss all aspects of turning over of services.

5.3 OUTGOING TURNOVER PERIOD
The Contractor shall be responsible for developing a turnover plan prior to the completion date of this contract, including option period if exercised. The turnover shall fully describe how the contractor will approach the removal of contractor property; any actions required to ensure continuity of operations by a successor contractor or the Government; and orderly transfer of contract responsibility to a successor contractor. The Contractor shall work with the successor contractor and the Government for a period beginning 30 days prior to the expiration of the contract to turnover this contract services and transition to the new contract services. The Contractor shall remove all contractor-owned equipment and supplies from the premises by the expiration/termination of the contract. The Contractor shall provide the turnover plan to the COR within 30 days of the COR's request.

6.0 SECURITY:

6.1 ACCESS TO GOVERNMENT FACILITIES:
Contractor shall possess a current **Top Secret/Sensitive Compartment Information (TS/SCI)** facilities clearance.

Access will be granted to Government facilities as required in support of the tasking. Contractor personnel performing work under this contract must meet all requirements for gaining access to US Government installations. The Contractor shall conform to all DOD, DON, and local (base/installation) security instructions; refer to Contractor Unclassified and Classified Access to Federally Controlled Facilities, Sensitive Information, Information Technology (IT) Systems or Protected Health Information.

All work is to be performed in accordance with DOD and Navy Operations Security (OPSEC) requirements and in accordance with the DD Form 254 and all attachments.

6.2 SECURITY CLEARANCES
Contractor personnel working on-site shall possess a **Top Secret/Sensitive Compartment Information (TS/SCI)** clearance as a condition of hire to support NIWDC services.

6.3 SECURITY PROCEDURES
The following special security procedures apply to administration and execution of this contract to ensure proper safeguarding of SCI:

- Personnel with an active TS/SCI with foreign family/association ties may require additional SCI processing through SSO Navy and/or DoD and may not be authorized to support this contract until that additional processing is complete. Due to the increased risk and potential delay in support, additional SCI processing shall be accomplished prior to contractor employees being assigned to support this requirement. Exceptions are granted on a case-by-case basis and require prior approval from the contract Contracting Officer’s Representative and agency Commanding Officer (COM).
- NIWDC Special Security Officer (SSO) will review the SCI requirements of this PWS prior to issuance of a DD Form 254.
- NIWDC Contracting Officer’s Representative (COR), SCI clearance required, will coordinate with the SSO on all requests by the Contractor for access, storage and processing of SCI materials and assist the Contractor in obtaining Government services and support from the SCI community, as required for contract performance.
- Contractor access to SCI shall be at Government facilities only.
- Contractors do not require substantial access to SCI.
- Contractors require entry and unescorted access to SCI areas.
- Use or storage of SCI at contractor operated facilities is not required.
- The Government contracting activity does not require access to SCI for contract award or oversight.

- Contractor will receive classified documents for reference only; however, if any classified information is generated in performance of this contract, it shall be derivatively classified and marked consistent with the source material.
- Contractor requires access to classified source data up to and including Top Secret in support of the work effort. Any extracts or use of such data requires the contractor to apply derivative classifications and markings consistent with the source documents. Use of "Multiple Sources" on the "Derived From" line necessitates compliance with the NISPOM, paragraph 4-208a, and the use of a bibliography.

7.0 MISCELLANEOUS

<u>7.1 MANDATORY CONTRACTOR TRAINING:</u>
Contractor will complete all annual training requirements identified by the Contracting Officer's Representative (COR).

<u>7.2 COMBATING TRAFFICKING IN PERSON (CTIP)</u>:
The Department of Defense (DOD) has a zero-tolerance policy regarding trafficking in persons. **FAR Subpart 22.17:** Prescribes overall federal regulation implementing 22 U.S.C. 7104 which applies to all acquisitions. Requires government contracts to (a) Prohibit contractors, contractor employees, subcontractors, and subcontractor employees from engaging in trafficking in persons during the period of performance of the contract. See FAR Provision 52.222-56 and FAR Clause 52.222-50.

<u>7.3 CONTRACTOR IDENTIFICATION:</u>
For all services provided under this PWS, the Contractor employee(s) shall identify themselves as Contractor personnel by introducing themselves or being introduced as Contractor personnel and displaying distinguishing badges or other visible identification for meetings with Government personnel. Additionally, contractor personnel shall identify themselves as Contractor employees in telephone conversations and in formal and informal written correspondence. The Contractor shall advise the Government on all changes in personnel (key or otherwise) during the performance of this effort. 30 days' notice will be given prior to personnel changes being made by the Contractor and emphasis should be made to thorough and complete turnover from one member to the other.

<u>7.4 GOVERNMENT FURNISHED INFORMATION:</u>
The Government will provide access to current information and data on a case-by-case basis. For permanent on-site personnel, government furnished material will be provided as necessary to perform any assigned task. All material provided shall be considered the property of the Government and will be returned to the designated representative or technical point of contact (TPOC) upon completion of the task. For permanent on-site contract personnel, the Government will provide a workspace and normal office equipment needed to complete the task. Contractor shall protect DoD sensitive unclassified data regardless of the location or ownership of the transport media, including, but not limited to mobile computing devices and removable storage media; whether Government furnished or contractor owned/leased. The Contractor shall comply with all current information assurance, information security policies, procedures, and

statutes applicable to DoD information technology; including the July 3, 2007 DoD CIO Policy Memorandum on Encryption of Sensitive Unclassified Data at Rest on Mobile Computing Devices and Removable Storage Media.